

Unsaturated and Saturated Flow in Response to Pumping of an Unconfined Aquifer: Numerical Investigation of Delayed Drainage

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Abstract

In this study, numerical methods are used to examine the capillary fringe hypothesis proposed by Nwankwor et al. (1992) to explain variations in the storage property of an unconfined aquifer during pumping. Based on field observations, the hypothesis suggested that the response of an unconfined aquifer to the stress imposed by pumping is largely controlled by the magnitude of vertical hydraulic gradients developed above the moving water table. After validation with a 1-D hysteretic laboratory experiment and a 2-D axisymmetric variably saturated semianalytical model, the numerical solution was used to simulate the pumping test of Nwankwor et al. (1992). Numerical parameters used for the simulation were chosen from experiments independent of the pumping test. There was good agreement between numerical results and field data at different depths and different radial distances from the pumping well. Numerically obtained variations in the magnitude of vertical gradients above the water table and the resulting variations in values of apparent specific yield also showed similar trends with the field data. The good agreement between the numerical and field results provides convincing evidence that the field observations and the resulting capillary fringe hypothesis concerning time variations of specific yield, are indeed consistent with the physical principles of variably saturated flow. The high storativity values frequently obtained by fitting early-time data to type curves were shown to correspond to the period of increasing hydraulic gradients, and may not be related to intrinsic properties of the aquifer, such as compressibility. In agreement with the "volume-balance method" of Nwankwor et al. (1984), it was also shown that the results confirm that the ultimate specific yield of an unconfined aquifer is more related to the drainable porosity than the low values often obtained from type-curve analysis.

Introduction

A major area of hydrogeologic endeavor has concerned the characterization of aquifers with respect to their storage and transmissive properties. These characteristics are important in assessing the potential of an aquifer as a water resource, and are also necessary for the prediction of both short-term and long-term responses of an aquifer to different hydraulic and environmental stresses. Furthermore, many procedures for rehabilitating contaminated aquifers require a good estimate of these parameters (Lehr and Nielsen, 1982; Gorelick and Voss, 1984). Common methods of obtaining in situ hydraulic characteristics of an aquifer include pumping tests. Though many factors can influence the reliability of such tests, a necessary prerequisite is that the theories upon which the analysis is made represent the processes that prevail during the tests.

Although there is a substantial technology available for analyzing the response of confined aquifers, development of adequate models for the analysis of unconfined aquifers has been a particularly difficult problem. The difficulty arises

because, as pumping proceeds and the water table declines, the thickness of the aquifer is not constant in time or in space and also because there are significant vertical hydraulic gradients, particularly at early times of pumping. As a consequence of these difficulties, there continues to be uncertainty concerning the adequacy of the assumptions that are normally made concerning the release of water from storage. The inflections often observed in the time-drawdown curves suggest that different physical processes dominate the flow domain at different times. This presents an interesting challenge for the development of interpretive models and procedures. One must first arrive at a physically correct explanation of the behavior, and this explanation must then be formulated mathematically such that the storage and transmissive properties of the aquifer can be determined.

Nwankwor et al. (1992) discussed the inadequacies in each of the two classes of mathematical formulations currently used to explain the response of unconfined aquifers to pumping. While Boulton's (1963) concept of delayed yield has hitherto been criticized because of the inclusion of an arbitrary delay index that lacks any physical meaning, the concept of instantaneous and complete release of water at the water table proposed by Neuman (1972) contradicts observable drainage processes (McWhorter and Sunada, 1977). There is an increasing amount of field and laboratory evidence to support the view that the unsaturated zone plays an important role in the delayed-yield process (Higgins, 1980; El-Kadi and Brutsaert, 1986; Brutsaert and El-Kadi, 1986).

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Received July 1991, revised February 1992, accepted March 1992.

Discussion open until May 1, 1993.

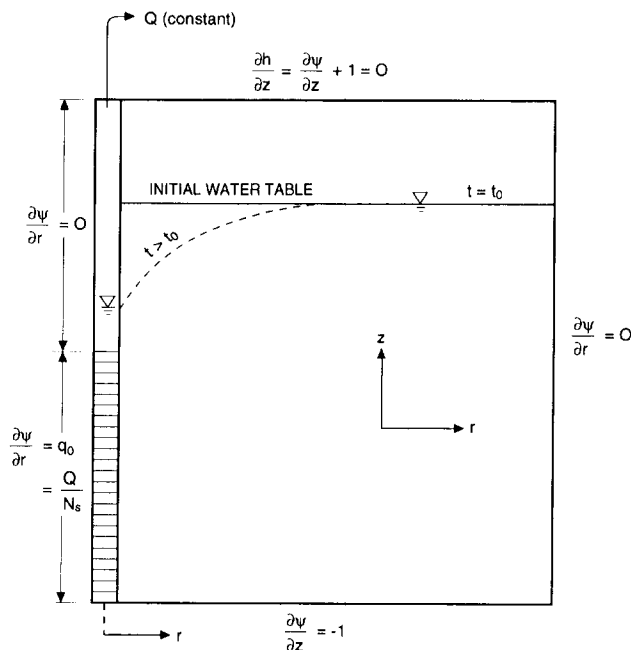


Fig. 1. A schematic of the problem domain and the boundary conditions.

Investigating the delayed-yield process, Nwankwor et al. (1984) reported the results of a detailed pumping test obtained from an unconfined sand aquifer located at Canadian Forces Base, Borden, Ontario, Canada. Piezometers were installed at different radial distances and depths across the saturated thickness of the aquifer. The detailed instrumentation presented a reasonable three-dimensional picture of the variation in hydraulic head within the aquifer as pumping proceeded. The specific yield of the aquifer material was obtained by three different methods:

1. from a laboratory drainage experiment, which indicated a specific-yield value of 0.3;
2. by the ratio of the cumulative discharge to the volume of the water-table drawdown cone (volume-balance method) estimated at different times since pumping started. This method gave values of specific yield that varied from 0.02 at 0.25 hours to 0.25 at 65 hours; and
3. by using type curves based on Boulton's model (Boulton, 1963; Prickett, 1965) and the Neuman (1975) model. Boulton's type curves gave a specific-yield value of 0.08 while Neuman's method gave a value of 0.07.

The authors concluded that both type-curve methods gave similar values of specific yield for the aquifer, but that the values were unrealistically low for a sand aquifer. The volume-balance method indicated that the specific yield increased as pumping proceeded and the values asymptotically approached that obtained from the laboratory drainage experiment. Because both the Boulton and Neuman models make simplifying assumptions about the complex processes at the water table, a possible explanation of the apparent discrepancies in specific-yield values is the inadequate representation of the drainage process.

Based on the field data collected around the moving water table during another pumping test, Nwankwor et al. (1992) presented a physically consistent explanation for

time-drawdown response of unconfined aquifers. The objective of this paper is to reproduce the trends in those field results by using an appropriate numerical method, thereby showing that the field data are in accordance with the physical principles of variably saturated flow. The results of numerical simulations will be explained in terms of the recently proposed capillary fringe hypothesis of Nwankwor et al. (1992) with a view to validating the conceptual model. We will also show that the volume-balance method does indeed represent the trends in apparent specific yield during pumping of unconfined aquifers. Although the numerical tests reported here were conducted using the physical and hydraulic properties of the Borden aquifer material, the generality of the results will later be extended to different aquifer materials in different hydrogeologic settings.

Description of the Numerical Model

Governing Equations and Problem Domain

The physical system simulated, including the appropriate boundary conditions, is shown in Figure 1. The governing equation for variably saturated radial flow in the aquifer is given by

$$\frac{\partial}{\partial r} \left(K_{rr} k_s(\psi) \frac{\partial \psi}{\partial r} \right) + \frac{K_{rr} k_s(\psi)}{r} \frac{\partial \psi}{\partial r} + \frac{\partial}{\partial z} \left[K_{zz} k_s(\psi) \left(\frac{\partial \psi}{\partial z} + 1 \right) \right] + q = [C(\psi) + S_w S_s] \frac{\partial \psi}{\partial t} \quad \dots (1)$$

where r, z = radial and vertical directions of coordinate axes, respectively [L]; $K_{ii}(i = r, z)$ = saturated principal hydraulic conductivity in direction i [LT^{-1}]; ψ = pressure head [L]; $k_s(\psi)$ = relative hydraulic conductivity [-]; $C(\psi)$ = specific moisture capacity [L^{-1}]; $S_w(\psi)$ = degree of saturation [-]; S_s = specific storage [L^{-1}]; t = time [T]; and q = a sink/source term [T^{-1}], i.e., the volume of water added to or extracted from a unit volume of the porous medium per unit time.

The problem domain consists of an unconfined aquifer underlain by an aquitard. A condition of no vertical flow is prescribed along the bottom boundary. The initial position of the water table is assumed to be sufficiently deep such that the water content at the surface approaches the residual value at all times. A condition of no vertical flow is thus prescribed for the upper boundary (ground surface). A condition of no flow is prescribed for the right boundary, to ensure that all the water pumped out of the aquifer originates from within the problem domain. The well pumps at a constant rate, Q ; it fully penetrates the aquifer with more than 60% of the initial saturated thickness screened, and it is considered to have both negligible storage and negligible radius. The left boundary is considered to be coincident with the axis of the pumping well. Being a line of symmetry, a condition of no horizontal flow is prescribed for this boundary except for the screened portion of the well where a uniform horizontal hydraulic gradient is assumed.

Because of the nonlinearity of equation (1) and the effects of hysteresis in the $k_s(\psi)$ and $C(\psi)$ relations, there is

no known analytical solution to the equation except for some special cases. Semianalytical solutions often neglect the effects of elastic storage of the aquifer and hysteresis. The numerical model SUNFLOW (Saturated UNsaturated FLOW) used for this study is based on the Galerkin finite-element formulation of equation (1). It uses an iterative procedure to handle the nonlinearity of the equation, and the sink or source term is expressed as the sum of the prescribed nodal fluxes q_i across the well screen, such that:

$$q = \sum_{i=1}^{N_s} q_i \quad (2)$$

where N_s is the total number of nodes on the prescribed flux boundary. SUNFLOW is an extension of the two-dimensional rectangular coordinate model described by Abdul et al. (1983) and Abdul (1985). The model incorporates the following features:

1. simulation of radial flow in both the saturated and the unsaturated zones of an unconfined aquifer;
2. full and partial penetration of production wells discharging at a specified rate;
3. anisotropy and heterogeneity of the aquifer material;
4. hysteresis of hydraulic parameters and compressibility of entrapped air, which may be important when modeling recovery or cyclic pumping;
5. problem formulation in either rectangular or cylindrical coordinate systems;
6. incorporation of seepage faces (pressure head is zero, but flux is outward from the boundary).

The structure of the original program was retained and the current model can be run in either 2-D Cartesian or radial coordinate systems. Hence, the detailed description of the original version as presented by Abdul et al. (1983) serves as a user's manual, and only a brief description of the program is given here.

Representation of Hydraulic and Storage Characteristics

The principal saturated hydraulic conductivities (K_{rr} and K_{zz}) are required input for the program as is the specific storage S_s , of the porous medium. The input parameters most difficult to incorporate in the equations are the relative hydraulic conductivity and the specific moisture capacity, which are hysteretic. Different workers have used different approaches to incorporate hysteresis into variably saturated flow models; Guitjens and Luthin (1971) used the method of independent domains; Gillham et al. (1976, 1979) used a hyperbolic cosine relationship. The method used in SUNFLOW is discussed briefly below.

According to van Genuchten (1980), the water content-pressure head relation can be represented by:

$$H = \frac{\theta - \theta_r}{\theta_s - \theta_r} = \left\{ \frac{1}{1 + (\alpha|\psi|)^l} \right\}^m \quad \psi \leq 0 \quad (3)$$

where H = dimensionless moisture content; θ = moisture content; θ_r = residual moisture content; θ_s = saturated

moisture content; α , l = curve-fitting parameters (α has a dimension L^{-1}); and

$$m = 1 - \frac{1}{l} \quad (0 < m < 1, l > 1) \quad (4)$$

In a recent study, Sakellariou-Makrantonaki et al. (1987) suggested that equation (3) is subject to the condition:

$$0.5 < m < 1 \quad (2 < l < \infty) \quad (5)$$

Values of l and m for the porous medium used in this study were found to be in agreement with the limits indicated in relation (5). By rearranging equation (3), the moisture content (θ) at any arbitrary pressure head is given by:

$$\theta(\psi) = \left\{ (\theta_s - \theta_r) \left(\frac{1}{1 + (\alpha|\psi|)^l} \right)^m \right\} + \theta_r \quad \psi \leq 0 \quad (6)$$

The degree of saturation S_w at any location can thus be evaluated by:

$$S_w = \frac{\theta(\psi)}{n} \quad (7)$$

where n is the porosity.

The specific moisture capacity is the slope of the moisture content-pressure head curve. It is obtained by differentiating equation (6) to give:

$$C(\psi) = \frac{d\theta}{d\psi} = \frac{-\alpha m (\theta_s - \theta_r)}{1 - m} H^{1/m} (1 - H^{1/m})^l \quad (8)$$

where the right-hand side is expressed in terms of the dimensionless moisture content, H , defined in (3). Once the sets of curve-fitting parameters that describe both the drainage and the wetting curves are defined, an infinite number of scanning curves can be constructed (Abdul, 1985).

Mualem (1976) proposed that the relative hydraulic conductivity can be represented by:

$$k_s(\psi) = \frac{\{1 - (\alpha|\psi|)^{l-1} [1 + (\alpha|\psi|)^l]^{-m}\}^2}{[1 + (\alpha|\psi|)^l]^{m/2}} \quad (9)$$

Using this approach, values of hydraulic conductivity can be calculated from the θ - ψ relation; hence measured values of hydraulic conductivity are not required. An alternative approach is based on the Averjanov-Irmay function (Lindsay, 1984; Abdul, 1985; El-Kadi and Brutsaert, 1986) by which the hydraulic conductivity is expressed as:

$$K(\theta) = a\theta^b \quad (10)$$

where $K(\theta)$ = measured hydraulic conductivity value [L/T]; and a , b = curve-fitting parameters (a has units of L/T). By this approach, a log-log plot of measured hydraulic conductivity against moisture content should give a straight line, and the graph must necessarily pass through the saturated hydraulic conductivity at saturated water content. The curve-fitting parameters (a and b) can respectively be obtained from the intercept and slope of the graph. An advantage of this approach is that values of a and b can easily be changed to study the sensitivity of model predictions to changes in the K - θ relation. This approach was also

found to give results that were in better agreement with experimental data, especially when the water content was close to saturation. Both options for representing $K(\theta)$ are included in SUNFLOW.

Method of Obtaining Curve-Fitting Soil Parameters

From equations (6) and (8), the hysteretic specific-moisture capacity can only be obtained if the curve-fitting parameters α and l are known. The parameter m can be calculated from l . Also, these are the only parameters required to obtain hydraulic conductivity if (9) is used. However, if equation (10) is used to describe hydraulic conductivity, parameters a and b are obtained by fitting the equation (10) to experimental data.

To obtain values of α and l , a Statistical Analysis System (SAS) computer program was used. The program, SAS PROC NLIN, carries out parameter estimation for nonlinear models through a Gauss-Newton minimization algorithm. A range of initial estimates must be supplied for the unknown parameters. The model to be tested [in this case, equation (6)] must also be supplied. The slope of the model with respect to each parameter to be determined should be supplied as well. These slopes are defined by:

$$\frac{d\theta}{dl} = -m(\theta_s - \theta_r)[\alpha|\psi|^l \ln(\alpha|\psi|)] \left[\frac{1}{1 + (\alpha|\psi|)^l} \right]^{m+1} \quad (11)$$

and

$$\frac{d\theta}{d\alpha} = -\frac{1}{\alpha} (\alpha|\psi|)^l (\theta_s - \theta_r) m \left[\frac{1}{1 + (\alpha|\psi|)^l} \right]^{m+1} \quad (12)$$

and when θ_r is also required by this method,

$$\frac{d\theta}{d\theta_r} = 1 - \left[\frac{1}{1 + (\alpha|\psi|)^l} \right]^m \quad (13)$$

The program uses the combination of the initial estimates to search for the grid points that give the lowest residual sum of squares. The program was used to determine the values of α , l , and θ_r for both drainage and wetting curves for the dune sand described by Gillham et al. (1976). For the drainage curve, the value of θ_r is 0.094 (read from the θ - ψ graph) while the program generated 0.09531. Similarly, for the wetting curve the program generated 0.10264 compared with 0.10 read from the graph. This method was deemed an acceptable way of obtaining the curve-fitting parameters from the measured retention characteristics of the aquifer material.

Model Validation

The radial coordinate formulation of SUNFLOW was tested against the experimental results of Gillham et al. (1976, 1979). In the experiment, a rectangular column, 0.128 m $\times$ 0.0565 m and 0.6 m high was packed with medium-fine dune sand. The column, initially saturated with water, was allowed to drain at a controlled rate for a length of time, after which it was rewetted. For the purpose of validation, SUNFLOW was used to simulate the experimental results

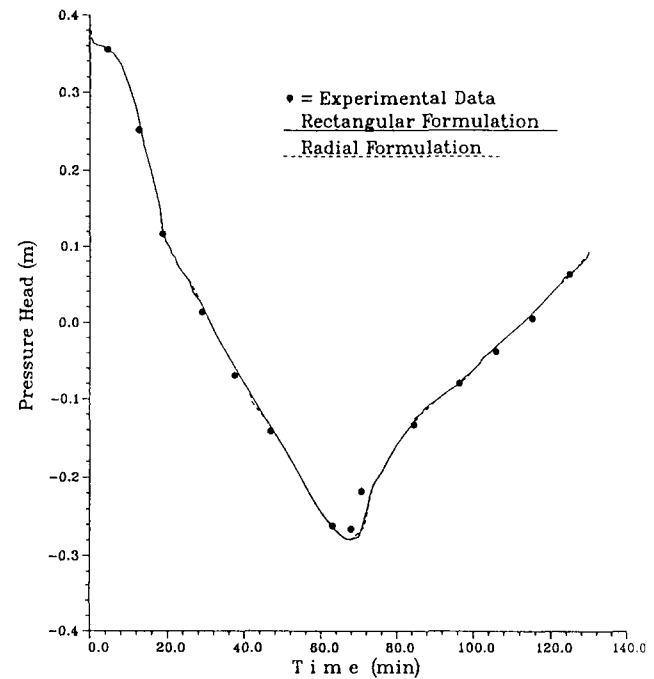


Fig. 2. Comparison of simulated and laboratory results for a one-dimensional drainage experiment.

using both the rectangular and the radial options of the program. The flow column was discretized into 120 triangular elements with $\Delta r = 0.024$ m, and $\Delta z = 0.02$ m. The curve-fitting parameters obtained for the soil characteristics are: $\alpha = 0.030$, $l = 7.44$ for the drainage curve; and $\alpha = 0.055$, $l = 4.77$ for the wetting curve. The results, as shown in Figure 2, indicate that the pressure-head profile generated by the radial option of SUNFLOW is identical to that obtained by Abdul (1985) (rectangular option). Both results are in excellent agreement with the experimental data.

The numerical model was also tested using the results of Kroszynski and Dagan (1975). It is emphasized here that no general solution to the variably saturated flow equation [equation (1)] is currently available for radial flow to a pumping well. The study of Kroszynski and Dagan (1975) includes an analytical solution for transient variably saturated flows with radial symmetry. The solution is approximate, however, and valid for the special case where soil characteristics can be represented by the simple analytical relationships:

$$\theta = \{(\theta_s - \theta_r) \exp[c(\psi - \psi_c)]\} + \theta_r \quad (14)$$

$$\text{and} \quad K(\theta) = K_{rr} \exp[c(\psi - \psi_c)] \quad (15)$$

where ψ_c = air entry value; and c = a parameter characterizing the unsaturated zone. Other parameters are as previously defined. It was possible to use the approximate solution to compare with the model of the current study because once θ_s , θ_r , c , and ψ_c are defined, equation (14) can be evaluated for different values of ψ to obtain the θ - ψ relation. Each value of ψ can then be substituted into equation (15) to obtain a corresponding value of K , thereby generating the K - θ relation.

Kroszynski and Dagan (1975) considered an unconfined aquifer, 13 m thick with an initial saturated thickness of 10 m. A screen of length 1.932 m was located 3.234 m below the static water table. The pumping well, having a diameter of 0.6 m discharged at a rate of 556 l/min (800 m³/day). Saturated principal hydraulic conductivities were $K_{rr} = 40$ m/day and $K_{zz} = 10$ m/day. Soil characteristics of the hypothetical aquifer material were obtained from equations (14) and (15), with $\psi_c = 0.35$ m; $c = 2.07$ m⁻¹; $\theta_s = 0.3$; and $\theta_r = 0.1$, such that $S_y = \theta_s - \theta_r = 0.2$. In the simulation, the external radius was set to 70 m. Certain modifications were made, mainly to ease the discretization of the problem domain. The length of the screen was changed to 1.9 m (instead of 1.93 m), and it was located 3.25 m (instead of 3.23 m) below the static water table. Also, the value of the characteristic parameter c used was 2 m⁻¹, rather than 2.07 m⁻¹. We believe that these minor modifications would have negligible effect on the results. To accommodate the fact that compressibility effects were neglected in the approximate analytical solution, specific storage was set to zero. Discretization commenced from a finite distance that corresponded to the well radius. This was considered necessary in order to minimize the problem of well storage which may not be negligible in a well of 0.6 m diameter at the specified pump-rate.

The comparison between predictions of the current numerical model with the semianalytical solution of Kroszynski and Dagan are summarized in Figure 3. The results are presented in the form of dimensionless drawdown (s') against dimensionless time (t') given by $s' = K_{rr}D's/Q$ and $t' = K_{zz}t/S_yD'$ where $D' =$ the initial saturated thickness of the aquifer [L]; and $s =$ drawdown [L]; and other variables are as previously defined. The figure also shows the results of Neuman's (1974) solution which considers flow

only in the saturated zone. Input parameters for the saturated flow model included a specific yield of 0.2 and specific storage of 10^{-8} m⁻¹, thereby eliminating compressibility effects.

Figure 3a shows the predictions of the three models for two locations; F is located at a depth of 0.25 m below the water table and at a radial distance of 54 m from the well, while G is located at a depth of 4 m below the water table and at a radial distance of 5.4 m from the well. There is excellent agreement between predictions of the three models for location G. Close to the water table (F), the drawdown values predicted by the saturated model are much lower than the corresponding predictions by the other two models that account for flow in the unsaturated zone. The predictions of the Kroszynski and Dagan analytical solution are almost identical to those of SUNFLOW for early and intermediate times. The higher values of drawdown predicted by SUNFLOW at late time is a reflection of the boundary condition imposed along the external boundary in the numerical model. The discretized domain extended to a radius of only 70 m from the pumping well; boundary conditions imposed along the external boundary would affect the results obtained from locations close to the boundary. Since a zero-flux condition was imposed along the boundary, the late-time drawdown predictions are expected to be artificially high.

Figure 3b shows the effect of radial distance on the time-drawdown curves. Locations C, D, and E are all 0.5 m below the static water table, but at radial distances of 5.4, 12, and 25.5 m, respectively. Agreement between the Kroszynski and Dagan (1975) model and SUNFLOW was reasonably good at all times and distances, though the agreement appeared to improve with greater distances from the pumping well. At all three locations, the saturated flow model predicted lower drawdown values at early times. The trend becomes reversed at late times for location C, nearest to the well. However, predictions of all three models are similar for late times at large values of radial distance (E).

The good agreement between the predictions of the numerical model of this study and the semianalytical solution of Kroszynski and Dagan (1975) supports the validity of SUNFLOW and its ability to simulate transient saturated-unsaturated flow under conditions appropriate for pumping of unconfined aquifers. Neuman's saturated flow model predicted drawdown values that are lower than those obtained from models that represent flow in both the saturated and unsaturated zones, particularly at early times of pumping. Differences between predictions of the two classes of models increase as the observation point moves closer to the water table. This observation is consistent with the deductions of Nwankwor et al. (1992) that Neuman's model underestimates drawdown of the water table.

Simulation of Pumping Test at the Borden Aquifer

The numerical model was used to simulate the results of the pumping reported in Nwankwor et al. (1984), and Nwankwor et al. (1992). The aquifer is 9 m thick, and is composed of "primarily horizontal, discontinuous lenses of

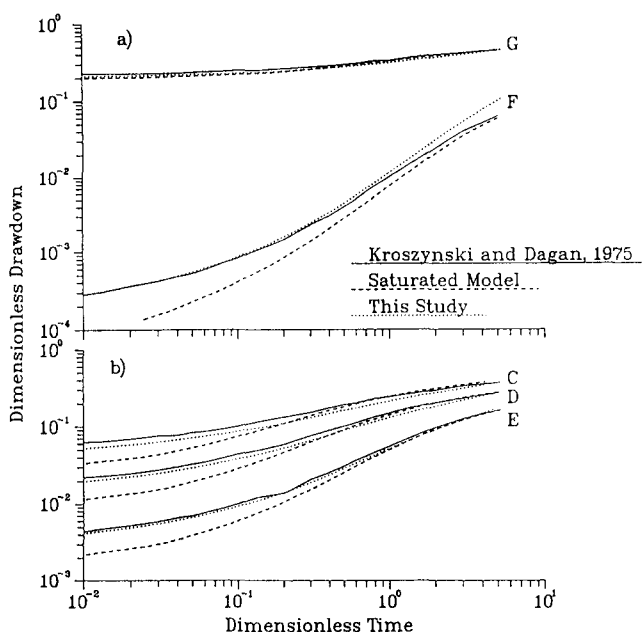


Fig. 3. (a) Effects of nearness of observation point to the static water table on simulated time-drawdown curves, and (b) effects of radial distance on simulated time-drawdown curves.

medium-grained, fine-grained and silty fine-grained sand" (Sudicky, 1986). The water table was located 2.3 m below the ground surface at the time the pumping test was conducted, although its position could have a seasonal variation of up to 1.0 m. A thick deposit of clayey silt underlies the aquifer. The pumping well has an inner diameter of 0.15 m with a 4 m screen located at the bottom of the aquifer. Data were collected from piezometers installed at different radial distances and terminated at different depths. Each piezometer was perforated over the lower 0.35 m. The test lasted for about 24 hours at a discharge rate of 60 l/min. Complete details of the instrumentation and test procedures are included in Nwankwor et al. (1984) and Nwankwor et al. (1992).

The domain was discretized starting 0.08 m from the axis of the pumping well. This was to make the simulation match the physical situation as closely as possible. Finer discretization was used close to the well, around the top of the screen and across the anticipated drawdown cone. It was also necessary to extend the fine discretization to some distance above the top of the capillary fringe in order to ensure that a reasonable number of nodes was specified within the zone where the magnitude of specific water capacity and hydraulic conductivity varied significantly with changes in pressure head. Running the model to 4000 minutes in test simulations, there was practically no difference in results obtained when the external radius was changed from 70 to 80 and 120 m. Hence, the external radius was kept at 80 m in subsequent simulations. Either a Dirichlet or Neuman condition could be specified for this boundary. However, a condition of zero horizontal flux was used to ensure that all the water pumped out of the aquifer originated from the discretized domain. This is particularly necessary to test the validity of Neuman's (1987) criticism of the volume-balance method. Parameters required by the numerical model were chosen in a manner that was independent of the pumping test. Only the value of principal hydraulic conductivity at saturation was adjusted to improve the fit between the field data and the numerical results. Table 1 gives the value of each parameter used in the simulation. The method by which the parameter values were determined is described below.

The curve-fitting parameters required to represent the θ - ψ relationship were obtained from the laboratory drainage experiment reported by Nwankwor et al. (1984). Figure 4 shows a good agreement between the laboratory data and the analytical model of van Genuchten (1980) using the best-fit values for the parameters α and l (Table 1). The porosity of the aquifer material (0.4) was obtained from the water-content profiles of Nwankwor et al. (1992).

The saturated principal hydraulic conductivity (K_{rr}) was obtained from results of the permeameter tests reported by Sudicky (1986) while K_{zz} , and hence anisotropy, was obtained from the relation:

$$K_{zz} = \frac{d}{\sum_{i=1}^j \frac{d_i}{K_i}} \quad (16)$$

Table 1. Hydraulic and Curve-Fitting Parameters Used in the Numerical Model to Characterize Borden Aquifer

Air entry value = 0.35 m

Saturated Hydraulic Conductivities:

Horizontal, $K_{rr} = 6.6 \times 10^{-5}$ m/s

Vertical, $K_{zz} = 4.2 \times 10^{-5}$ m/s

Anisotropy, $K_{zz}/K_{rr} = 0.64$

Storage Parameters:

Saturated water content, $\theta_s = 0.37$

Residual water content, $\theta_r = 0.07$

Specific yield, $S_y = \theta_s - \theta_r = 0.30$

Specific storage, $S_s = 3.25 \times 10^{-4} \text{ m}^{-1}$

Curve-Fitting Parameters:

α and l in Mualem's model

6.095 and 0.019 (drainage)

4.21 and 0.032 (wetting)

a and b in equation (10)

43.17 and 4.72

where d is the length of the soil column for which hydraulic conductivity (K_i) was determined; d_i is the thickness of the layer; and j is the total number of layers used for evaluating K_{zz} . Parameters needed to represent the K - θ relation as given in equation (10) were obtained from the laboratory experiment of Abdul (1985). In the present study, the best fit to the experimental data was obtained by increasing the saturated hydraulic conductivity to 6.6×10^{-5} m/s. This is slightly greater than the saturated value used by Abdul (5.0×10^{-5} m/s) and a factor of about three lower than that

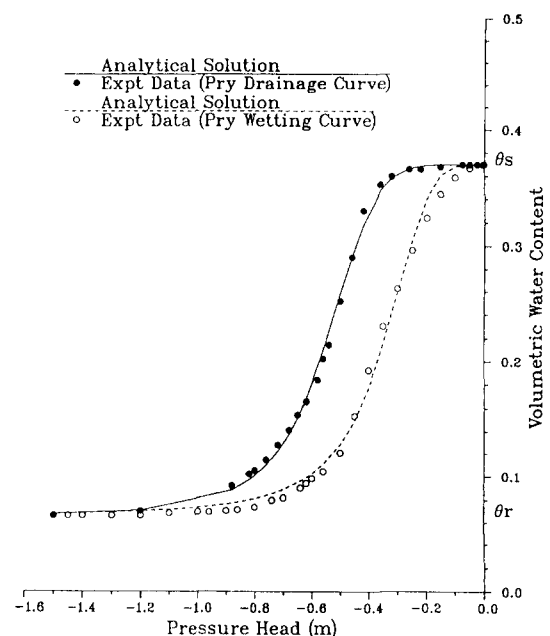


Fig. 4. Drainage and wetting curves for Borden sand. Experimental data (symbols) and analytical model of van Genuchten (1980) for α and $l = 6.095, 0.019$ (drainage); $4.242, 0.032$ (wetting).

obtained by Nwankwor (1984) by fitting time-drawdown data to type curves (1.4 to 2.2×10^{-4} m/s). This adjusted value used in the simulations of this study is well within the range measured using slug tests (Nwankwor et al., 1984) and grain-size correlations (MacFarlane et al., 1983). As shown in Figure 5, the adjusted K - θ function used in this study was very similar to that reported in Abdul (1985).

Specific storage of unconfined aquifers is commonly determined by fitting type curves to early time-drawdown data. This procedure frequently gives values that are up to an order of magnitude greater than those determined for confined aquifers. Neuman (1972, 1974, 1979, 1981) suggests that compressibility of unconfined aquifers may be greater than that of similar confined aquifers. Nwankwor et al. (1992), on the other hand, suggest that type-curve values are artificially high as a consequence of pure drainage at early time. For the purpose of this study, specific storage was evaluated using the relationship:

$$S_s = \rho g(\varphi + n\beta) \quad (17)$$

where ρ = density of water [M/L^3]; g = acceleration due to gravity [L/T^2]; φ = compressibility of the porous medium [LT^2/M]; β = compressibility of water [LT^2/M]; and n = porosity [$/$]. Freeze and Cherry (1979) gave the value of β , and estimates of φ for different geologic materials. The value of α used in this study was 3.3×10^{-8} m²/N which is within the range of 10^{-7} and 10^{-9} m²/N suggested for sands. The resulting specific storage is much lower than the average value Nwankwor et al. (1984) obtained by type-curve method (6.0×10^{-4} m⁻¹).

Results and Discussion

Simulated and measured time-drawdown graphs were compared at radial distances of 5 and 15 m from the pumping well (Figure 6). At each radial distance, the results were compared for two depths, just below the static water table and 2 m from the bottom of the aquifer. For the period up to 24 hours, the measured data of Nwankwor et al. (1992) were used for comparison with the simulated data. Those of Nwankwor et al. (1984) were used to extend the field data beyond 24 hours for late-time comparison because the same pumping rate was used for both experiments, and both sets of data showed identical trends in the 24 hour duration of the recent test. Results obtained using the Neuman (1974) analytical solution are also included on each graph. Neuman (1974) solutions are included for two values of specific yield, 0.05 as obtained from type-curve analysis (Nwankwor et al., 1984) and 0.3 which is believed to be the drainable porosity of the aquifer material as obtained from $S_y = \theta_s - \theta_r$.

There is good agreement between the field data and the results of the numerical simulations at all times, particularly for the point located 2 m above the aquitard [Figures 6(b) and (d)]. Although the late-time field data are slightly lower than the numerical results at a radial distance of 5 m [Figures 6(a) and (b)], the general trends of both sets of data are in agreement at all locations. No field data were available at early time for the water-table piezometer located at 15 m [Figure 6(c)]. This is because drawdown values were so

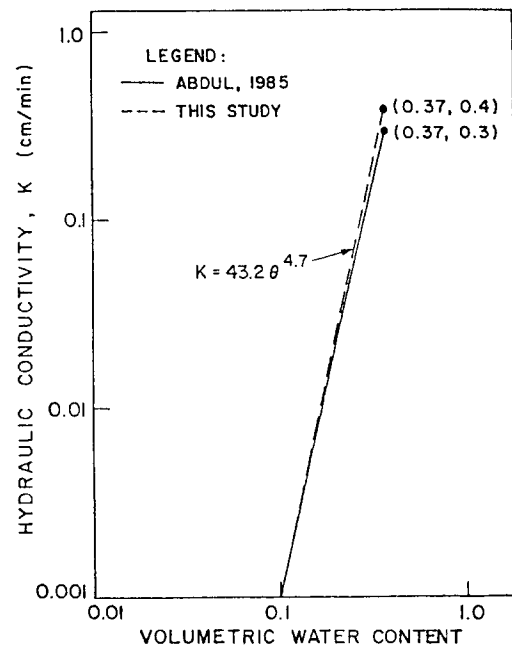


Fig. 5. Hydraulic conductivity-water content relation for Borden sand.

small that they could not be measured accurately. The questionable agreement between the field data and the numerical results shown in Figure 6(c) is inconsistent with the bulk of the data and may be the results of uncertainties in the field measurements at this location or could be caused by local heterogeneities in the vicinity of the piezometer.

For all locations, the saturated model with a specific yield of 0.3 gives results that are in excellent agreement with

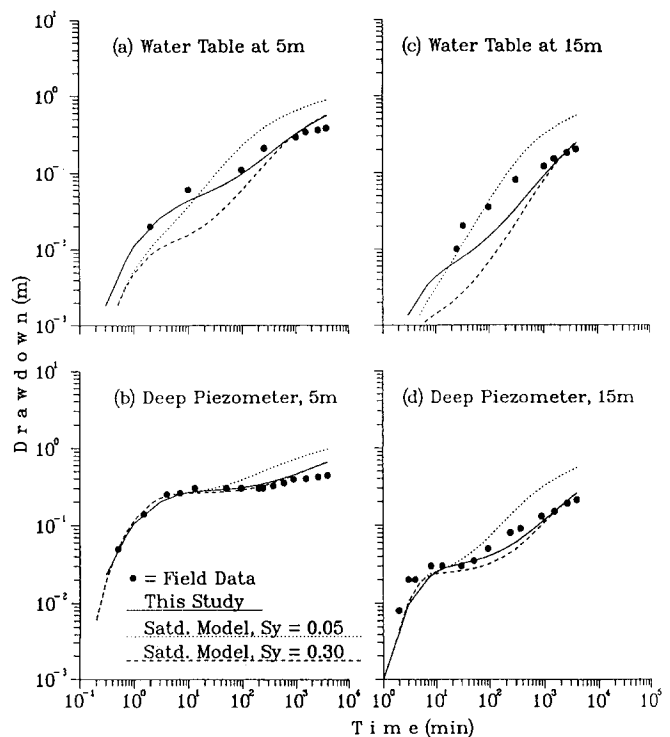


Fig. 6. Time-drawdown curves at selected radial distances from the pumping well and at different depth.

the numerical results (and hence the field data) at late times. Using a specific yield of 0.05, the Neuman solution overpredicts late-time drawdowns at all locations. While the time-drawdown responses of the deep piezometers [Figures 6(b) and (d)] are adequately predicted with a specific yield of 0.3, the Neuman solution did not give a good prediction of the water table [Figures 6(a) and (c)]. A specific-yield value of 0.3 underpredicts the early-time and intermediate-time responses of the water table, while the low specific-yield value of 0.05 gives a trend that is quite different from the water-table response as indicated by both the field data and the numerical results. With a specific-yield value of 0.05, the Neuman solution underpredicts drawdown at early times and overpredicts drawdown at intermediate and late times, at both radial distances from the pumping well.

The good agreement between the numerical results and field data suggests that SUNFLOW adequately represents the trends in the time-drawdown relationships as observed in the field study. Hydraulic and storage parameters used for the simulations are therefore representative of the aquifer. This includes a specific storage determined from the fundamental elastic properties of the aquifer material without recourse to an artificial increase in its magnitude, while the component of gravity drainage was determined by the specific water capacity (the slope of the water content-pressure head relation) of the aquifer material. It is, therefore, considered appropriate to use the numerical model to investigate the physical processes associated with the aquifer response to pumping.

Evaluation of Capillary Fringe Hypothesis

Having tested SUNFLOW with the Borden pumping test, it was used to evaluate the explanation given by Nwankwor et al. (1992) for the characteristic slope of time-drawdown curves for unconfined aquifers. In particular, it was shown that an extension of capillary fringe during pumping resulted in excess storage of water above the declining water table. The excess storage was described to be a necessary consequence of the rotation of the pressure head profile in response to changes in the magnitude of vertical hydraulic gradients across the water table as pumping progressed. Figures 7a, 7b, and 7c show the numerically evaluated profiles of pressure head versus height above the moving water table as established at selected times and at radial distances of 1, 5, and 15 m, respectively. The condition of static equilibrium is also represented on each figure. Considering the drainage curve for the aquifer material (Figure 4), the capillary fringe is expected to extend to a height of about 0.30 m above the water table under the condition of static equilibrium. Also, the water content is expected to approach the residual value at locations about 0.80 m above the water table and beyond. Hence, the results are evaluated for locations within a height of 0.8 m above the moving water table.

At a radial distance of 1 m (Figure 7a), there is a rotation of the pressure head profile away from the position of static equilibrium between the commencement of pumping and 25 minutes. The direction of rotation suggests that the vertical hydraulic gradients are directed downwards,

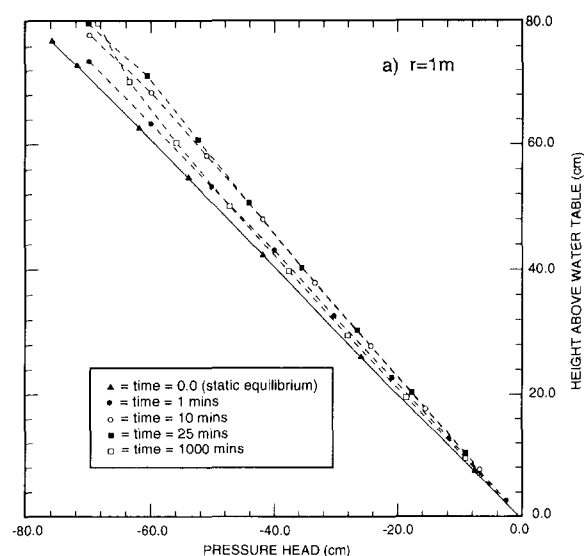


Fig. 7a. Variations in pressure head depth profiles at some selected times since pumping started; numerical results at 1 m from the pumping well.

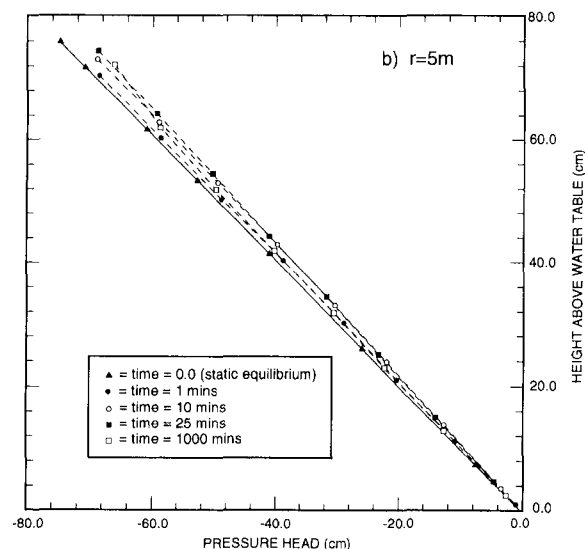


Fig. 7b. Variations in pressure head depth profiles at some selected times since pumping started; numerical results at 5 m from the pumping well.

towards the base of the aquifer. At later time (1000 minutes in Figure 7a), the profile again approaches the position of static equilibrium. The sequence of rotation indicated by the numerical model is in agreement with the sequence observed in the field study, though the magnitude of the simulated rotation was somewhat less than the measured values. The same sequence of rotation also can be identified at radial distances of 5 and 15 m (Figures 7b and 7c, respectively) with the magnitude of rotation decreasing as the radial distance from the pumping well increases. The times presented in Figures 7a, 7b, and 7c were selected only to indicate the sequence and directions of rotations; the times may not necessarily correspond to periods at which the highest or the lowest vertical hydraulic gradients were attained at each radial distance. However, it is reasonable to suggest that, 1

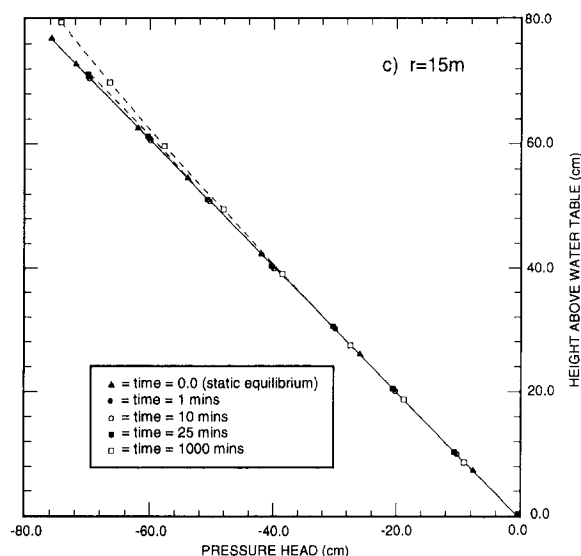


Fig. 7c. Variations in pressure head depth profiles at some selected times since pumping started; numerical results at 15 m from the pumping well.

m from the pumping well, maximum rotation of the pressure head profile occurred sometime between 10 and 25 minutes.

Figures 8a, 8b, and 8c show simulated variations in water content versus height above the water table at the selected times and radial distances corresponding to those plotted in Figures 7a, 7b, and 7c. The condition of static equilibrium represented on each figure is identical to the drainage curve of Figure 4 rotated through 90° in the clockwise direction. Figure 8a shows that the capillary fringe expands at early time of pumping and contracts back towards its initial thickness at late times. At time $t = 1000$ minutes, the profile has almost completely reverted back to the initial state. Furthermore, Figure 8a clearly shows the

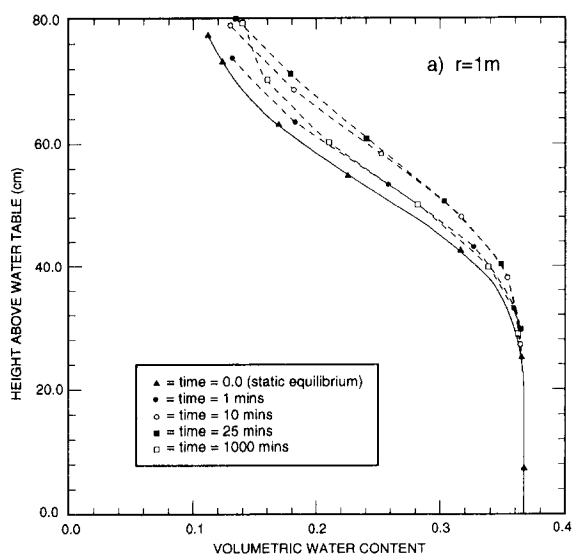


Fig. 8a. Variations in water content depth profiles at some selected times since pumping started; numerical results at 1 m from the pumping well.

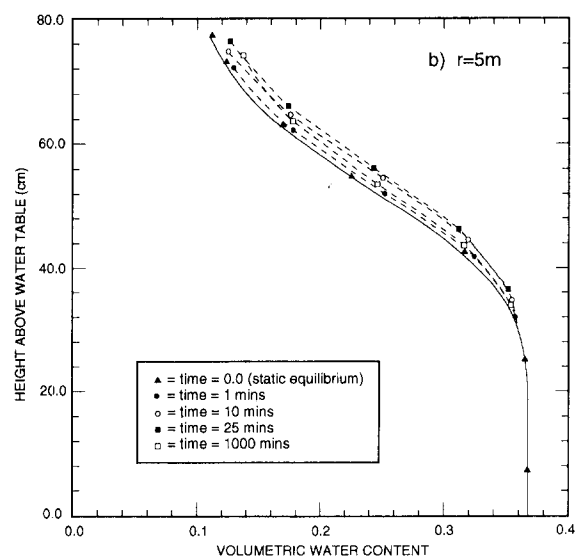


Fig. 8b. Variations in water content depth profiles at some selected times since pumping started; numerical results at 5 m from the pumping well.

development of excess storage above the water table as the water content profile stretches at early times of pumping. At 5 and 15 m (Figures 8b and 8c, respectively) from the pumping well, a sequence similar to that plotted in Figure 8a is also identified. Figures 8a, 8b, and 8c show that the magnitude of the observed variations decreases as radial distance from the pumping well increases.

Variations in water content and pressure head profiles suggest that the capillary fringe stretches to attain a maximum thickness and later decreases towards its initial value. Investigating this further, the thickness of the capillary fringe was evaluated at different times and different radial distances. The results are presented in Figure 9 as plots of percent extension of the capillary fringe versus time. The top

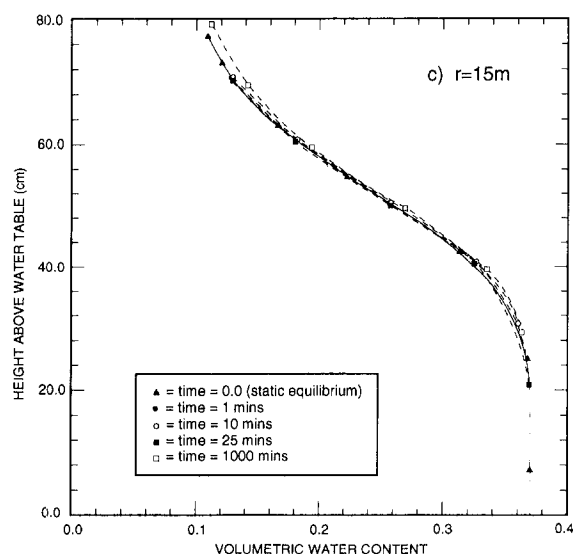


Fig. 8c. Variations in water content depth profiles at some selected times since pumping started; numerical results at 15 m from the pumping well.

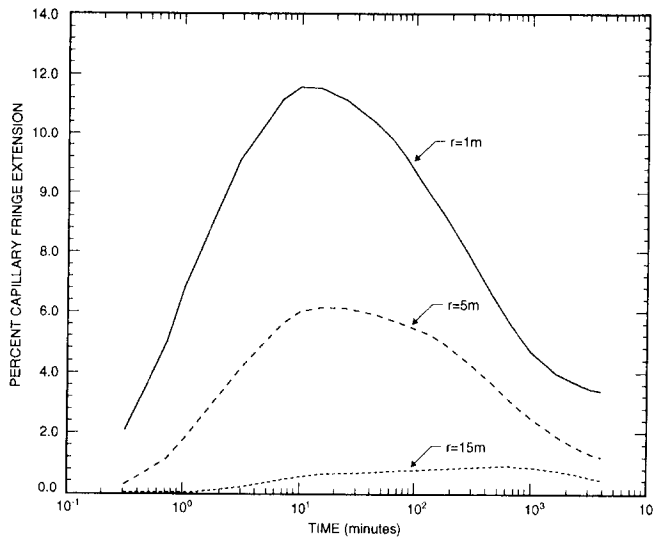


Fig. 9. Simulated variations in the thickness of the capillary fringe as pumping progressed.

of the capillary fringe was defined as the surface where the pressure head became numerically equal to the negative of the air entry value. Figure 9 clearly shows that the thickness of the capillary fringe increases towards a maximum at early times of pumping, and decreases towards the initial state at intermediate and late times. It also indicates that the expansion of the capillary fringe is greatest in areas closest to the pumping well. These results show trends that are similar to the field data of Nwankwor et al. (1992) and are consistent with the capillary fringe hypothesis put forward by the authors.

Apparent Specific Yield

When water is extracted from an unconfined aquifer, the apparent specific yield (S_{ay}) is expressed as the ratio of the volume of water removed to the resulting change in the volume of the drawdown cone. Thus,

$$S_{ay} = \frac{\Delta V_w}{\Delta V_c} \quad (18)$$

where ΔV_w = volume of water pumped out [L^3]; and ΔV_c = resulting change in the volume of the drawdown cone [L^3]. Considering the radial geometry of the problem, the denominator can be replaced by the expression

$$\Delta V_c = 2\pi \int_{r_w}^{\infty} (sr) dr \quad (19)$$

where s = drawdown of the water table [L]; r_w = the well radius [L]; and r = radial distance from the pumping well [L]. Under field conditions, the upper boundary of integration is invariably replaced by $r = r_m$ where r_m is the radial distance at which there is no measurable drawdown. The choice of r_m in field studies is limited by the sensitivity of the measuring device. It is doubtful if measurements could be accurate beyond a few millimeters. This limitation exposed the results of Nwankwor et al. (1984) to the comment of Neuman (1987) who argued that the volume-balance

method may not account for some water derived from areas outside the observable cone of depression.

In this study, apparent specific yield was evaluated using the trapezoidal integration method rather than the graphical method used by Nwankwor et al. (1984). Also, setting the external boundary to a zero flux boundary ensures that no water leaves or enters the problem domain. The results can be discussed in relation to equation (18). Since discharge was maintained at a constant rate, ΔV_w remains the same for equal time intervals. Hence, any variation in apparent specific yield is a reflection of variations in ΔV_c . A low value of apparent specific yield indicates a high value of cumulative drawdown and corresponds to a rapid change in the position of the water table (or top of the capillary fringe). Using the volume-balance method, Nwankwor et al. (1984) observed an asymptotic increase of apparent specific yield towards the drainable porosity.

As shown in Figure 10, the trend in the numerical result is very similar to that determined from the field observations. The lower values of apparent specific yield obtained from the field study, compared to those of the numerical experiments, are probably the result of the sensitivity of the devices used to measure drawdown and thus uncertainty in the outer limit of integration (r_m). Since the drainable porosity of the aquifer material is 0.3 ($\theta_s - \theta_r$) in Figure 4, the numerical experiments further confirm that the late-time storage property of the Borden aquifer material is closer to 0.3 than the low values obtained from type-curve analysis. This behavior is consistent with pressure head and water content profiles discussed in Nwankwor et al. (1992). The figure also shows that the early- and intermediate-time values of apparent specific yield obtained with respect to the top of the capillary fringe are greater than similar results obtained with respect to the water table. The results there-

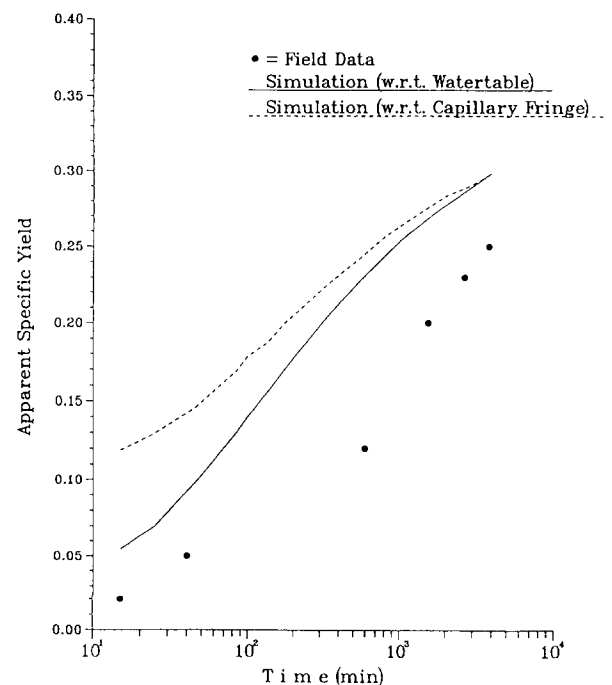


Fig. 10. Variation of apparent specific yield with time.

fore agree with the suggestion that higher values of specific yield would be obtained if drawdown measurements were made with respect to the top of the capillary fringe rather than the water table. In accordance with the capillary fringe hypothesis, both sets of results are similar at late times, when the magnitude of vertical hydraulic gradients become very small.

Conclusion

Considering the good agreement between the results of the numerical simulations and those of the semianalytical solution of Kroszynski and Dagan (1975), the numerical model (SUNFLOW) gave acceptable representation of saturated-unsaturated flow processes. The numerical results were in good agreement with the field data. Interpretations of the simulations were consistent with the observations and explanations of Nwankwor et al. (1992) in relation to the variations in the storage parameters of unconfined aquifers. The observations and explanations are therefore consistent with the current understanding of flow in variably saturated porous media. In agreement with the conceptual model proposed by Nwankwor et al. (1992), the results of numerical simulations confirmed that the early-time behavior of an unconfined aquifer is mainly the result of increasing vertical hydraulic gradients, leading to an extension of both the capillary fringe and the profile of water content versus height above the water table. Consequently, excess storage occurs above the water table. At intermediate times of pumping, the magnitude of vertical gradients declines, and the excess storage above the water table is gradually released as the capillary fringe and the water content profile contract. The time-drawdown curve thus indicates a decrease in the rate of drawdown. At late times, vertical hydraulic gradients are negligible, the effect of the capillary fringe becomes minimal, and the effective specific yield therefore approaches the drainable porosity.

Agreement between the late-time results of the numerical simulations and the Neuman model with a specific yield of 0.3 suggests that a value close to the drainable porosity of the aquifer material would be obtained from type-curve analysis if pumping tests were conducted for a sufficiently long period. Hydrologists frequently try to fit field data to type curves in the region around the point of inflection. While it is the region where there is marked sensitivity to curve-fitting parameters (β on Boulton curves and σ on Neuman curves), it is also the region where there is still appreciable excess storage above the water table. Depending on aquifer characteristics, and distances from both the pumping well and the water table, this procedure would give a specific yield value that is much lower than the drainable porosity because the effective specific yield at times corresponding to the inflection point could be much lower than the drainable porosity. The inflection region for the Borden sand occurred after pumping for about two hours. Both field data and the numerical results indicate that the effective specific yield at two hours is about 0.08, which is close to the value obtained by type curves.

There was no need to arbitrarily increase the aquifer compressibility (by orders of magnitude) before the numeri-

cal results matched the early-time field data. Therefore, the suggestion that unconfined aquifers have much higher compressibility values than similar confined aquifers appears to be incorrect. In fact, though not shown, an increase in aquifer compressibility moved the numerical results further away from field data. A similar observation was made recently by Brutsaert and El-Kadi (1986).

Acknowledgment

The authors wish to thank J. A. Cherry, E. O. Frind, and E. A. Sudicky of Waterloo Centre for Groundwater Research for valuable suggestions at different stages of this study. Discussions with D. McWhorter of Colorado State University were also helpful. Funding for the study was provided through NSERC Operating Grant No. 6414 and by the Federal Ministry of Water Resources, Nigeria in the form of a scholarship to the senior author.

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